

The GALACTIC CENTRE: Best Images Ever First Results from SHARP at the NTT At the Max-Planck-Institut a System for High Angular Resolution Pictures, SHARP, has been developed during the last 18 months for observations in the near infrared spectral range (1 to 2.5 μm). It is based on a 256 x 256 HgCdTe NICMOS3 array manufactured by Rockwell Inc. The camera has an image scale of 0.05 arcsec/pixel at the Nasmyth focus of the ESO 3.5-m New Technology Telescope. The electronics and the data acquisition system allow the recording of frame rates up to 10 Hz for speckle observations with a built-in cold shutter. Appropriate software has been developed for on-line quick-look data reduction (long exposure, shift-and-add, etc.). The system sensitivity at 1 Hz data rate allows detection of $K = 9.5$ in 1 arcsec seeing and fainter for better seeing. The first observing run took place between August 18 and 23, 1991. The figure shows a K-band image of the inner region of the galactic centre (6.4×6.4 arcsec corresponding to 0.25×0.25 pc, North is up and East to the right; a scale of 1 arcsec is marked). The image is a result of 1000 frames with 0.5 sec and 1 sec exposure time per image using the shift-and-add method and ten iterations of the Lucy image sharpening algorithm. The average instantaneous seeing was of the order of 0.4 arcsec and the resulting final resolution in the image is 0.25 arcsec. In the combination of spatial resolution and sensitivity this image by far surpasses anything available till now. The basic new results on the structure of the near infrared emission of the central 0.25 parsec of the Galaxy emerge already at this early stage of analysis. First, the IRS16 complex is resolved into about 15 compact sources, most of which may be hot massive stars. Identifications are marked in the figure. Second, we find from repeated exposures a $K = 12.5$ object within 0.2 arcsec of the radio source SgrA*, whose location and positional uncertainty are marked by the cross in the figure. This source may represent the long sought-for infrared counterpart of the compact radio source. We would like to thank the ESO Director General for his vision to admit SHARP at the NTT and the ESO staff at Garching and on La Silla for their excellent professional support and enthusiastic commitment. Will La Silla Succumb to the VLT? Concern has spread around the ESO community about the future of La Silla. Seeing the great technical efforts required by the VLT, some people fear that, according to the law of "man-power conservation", the efforts at La Silla will be correspondingly

reduced, causing in due time a deterioration of its present quality and diversity. Is this apprehension based on real-life experience or rather on expectations? An informal round-table discussion took place at La Silla on this subject in mid-August. The discussion started with an analysis of the present situation on the mountain, trying to single out its weak points on the basis of the outcome of the Users Committee Meeting held last May. A general consensus expressed concern about the present status of detectors, both optical and infrared, some of which appear to be out of date. The causes of this relatively negative situation were ascribed to the rather large number of CCDs now in operation at La Silla, probably more than at any other observatory in the world, and into the difficulty, at least till a few years ago, of getting modern IR detectors, due to export license problems. The Director General, responding to a somewhat pessimistic view about the rate of improvement, promised that major efforts will be spent at the ESO Headquarters to replace as fast as possible the bad detectors; in the IR in particular, a Rockwell 256x 256 array that is believed to be one of the best on the market, will be ready at the telescope early next year. The coming into operation of the NTT has been unanimously reported as a small-scale example of the impact the VLT might have on La Silla operations. In particular, the Director General lamented the underestimation of the amount of work required at La Silla to make this new telescope ready for common users: three years of heavy work have been necessary. In general it has to be carefully considered that whenever an instrument has been finished in Garching and is delivered at the telescope, a non-negligible amount of work is still required at the Chilean site to take care of all those more or less important details which have been overlooked. Some of those present noted that the NTT has also absorbed part of the resources of the Astronomy Support Department, and this has resulted in less assistance at telescopes like the 1.5-m Danish or the 2.2-m. Due to the pressure of the VLT project, the Director General reported 24 positions at La Silla will be phased out of the existing 140 over the next two and a half years. This reduction will affect proportionally the various departments and, to maintain and improve the La Silla standards, a process of "streamlining" will be necessary. An important part of the staff will be moved to Santiago, with the aim of reducing the number of people on

the mountain and to economically support both La Silla and VLT Observatories in the future. Fewer people on the mountain imply less board and lodging, less transport, fewer administrative requirements, allowing some economies. This of course involves a certain amount of simplification of La Silla: telescopes, like the Bochum and perhaps the ESO 50-cm, will be closed, the number of instrument change-overs will be reduced, the scheduled runs will become longer. Less direct assistance to the astronomers will necessarily imply the preparation of better and more detailed telescope and instrument manuals. The Director General illustrated the upgrades already taking place or planned for the next years: the 1.5-m Danish adapter will be renewed; the 90-cm Dutch has a new adapter and a CCD camera; IRAC will be upgraded; the prime focus at the 3.6-m is available for direct imaging (which could then be removed from the 2.2-m); the fibre spectrograph MEFOS will also be installed at the 3.6-m. A possible evolution of the telescopes and auxiliary equipment from now to 1993 and 1996 is shown in three ESO menus, according to a document presented by the Director General to the Council. These are illustrations, not, as yet, decisions. However, the simplification of the La Silla instrumentation, as remarked by the Director General, is a very delicate process, especially when the final decisions about the various instruments have to be taken. Many people, for example, would be upset if the B&C spectrograph, the only one allowing certain investigations of galaxy dynamics, is removed from the 2.2-m, others will cry if CASPEC is confined to the ESO 1.5-m, and even disagrees with the Director General about removing the direct imaging facility at the 2.2-m. It is obvious that a larger involvement of the community of users in these decisions must be ensured. The importance of playing a leading role in large-scale digitized surveys of the sky was emphasized, to avoid to be confined to a secondary place as may have happened with the ESO Schmidt compared to the UK Schmidt. The Director General replied that in this respect the scientific pressure of the ESO astronomers will be the driving factor. A general consensus was expressed about the importance of transmitting the experience gained in the VLT design to La Silla and to ensure a strong coupling of this project with the La Silla management. The VLT standards will become the La Silla standards: ethernet, VME, UNIX, MIDAS, VxWorks. The people of the VLT project will spend a considerable fraction

of their time in Chile: having many short duty trips would simply not work in the case of the VLT, and this will allow the interchange of precious experience. Finally, concerns were expressed about the future of MIDAS, the standard image-processing system at the telescopes in the VLT era. As remarked also in the Users Committee last May, although there is a general satisfaction with its improvements, the progress of MIDAS is considered slow compared to IRAF. This fact together with the improvement of reduction facilities at the home institutes, is the probable cause of the decrease in the number of astronomers coming to Garching to use the ESO facilities. The Director General explained that it is not ESO's intention to compete with the much larger manpower available for the development of IRAF/STSDAS. MIDAS will be offered as built-in-house software for off-line reductions of data acquired with ESO telescopes and, with the same core, but with different and more user-friendly interfaces, as data acquisition software. MIDAS is coming into La Silla telescopes' control rooms! This was more or less the point reached when the round table had to cede to an incipient clear observing night. I hope that the reader could get from this incomplete report enough material to answer the primitive question indicated by the title of this contribution.

A Survey of Nearby Clusters of Galaxies Introduction

Detailed analysis of the structure of a large sample (well representative of all richness and morphological classes) of rich clusters of galaxies may provide several constraints for theories of the formation of large-scale structure on scales of the order of a few Mpc. Since the overall relaxation time of a cluster is larger than the Hubble time, it could be expected that present-day clusters are at best partially relaxed -a situation which could be revealed by the presence of subclustering. The convincing detection of dynamically significant substructure in clusters may put severe constraints on the scenarios of formation and on the shape of the initial fluctuation spectrum. However, this detection, as well as dynamical analysis, requires a large number (at least 100) redshifts per cluster. Peculiar velocities of clusters with respect to the Hubble flow could in principle reveal the characteristics of the mass distribution on larger scales (such as the inter-cluster distance). The Cold Dark Matter Scenario, however, does not predict large-scale peculiar velocities with respect to the Hubble flow for rich clusters. It therefore appears desirable to confirm e.g. in the Southern

galactic hemisphere the large-scale peculiar motions which have been claimed to exist but questioned by other authors. Again, this requires a suitable sample and several redshifts per cluster.

Definition of an Observational Programme

The preceding motivations have led us to design an observational programme which has been accepted by ESO as a Key Programme. We have thus selected a sample of clusters of galaxies from the revised and South extended Abell catalogue; the sample can actually be divided into two subsamples since the selection has been performed following two different criteria. On the one side we needed a set of clusters sufficiently rich to allow a meaningful structural and dynamical analysis, possibly covering almost equally all Bautz-Morgan morphological types and having a redshift of $z \leq 0.05$. This last requirement ensures that we explore several core radii and cover a sufficiently large range of magnitudes on a field of 30 arcmin (Optopus field). We found 30 clusters (the so-called "structure" clusters) satisfying all these requirements. On the other side, we selected 100 clusters (the "peculiar" clusters) which are expected to form a complete sample up to $z=0.1$ in the Southern galactic hemisphere. This large subsample will be used to map and analyse the large-scale peculiar motion field. There is a partial overlap (12 objects) among the two subsamples, giving a total of 118 clusters for the whole sample. We plan to obtain spectroscopy and photometry for about 150 members of the "structure" clusters and 30 to 50 members of the "peculiar" ones. Spectroscopy is being obtained by using the OPTOPUS instrument at the ESO 3.60-m, whilst CCD photometry (Danish 1.54-m) is required in order to calibrate our photographic one. For each selected cluster, catalogues of galaxies have been obtained by scanning photographic plates (glass copies of the red PSS for the northern clusters and film copies of the SRC survey for the southern ones) using the Leiden ASTROSCAN plate-measuring machine. Objects are selected around the cluster centre within several Abell radii. The typical size of a scanning region is about one square degree for "peculiar motion cluster" and 4 square degrees for "structure" ones. By setting a threshold value (typically at 5 times the sky noise above the background) a list of objects is then produced. For each object the following final parameters are kept: (i) the centre of gravity position, (ii) a photometric parameter, the logarithm of which is in good approximation a linear

function of magnitude. (iii) the second moment of the density distribution, which is a measure of the size of the image, and (iv) the number of image pixels which have a photographic density above the detection threshold. Star-galaxy separation is performed by plotting: second moment or the density versus photometric parameter. The stars' locus is well defined and separated from the galaxy one. Anyway, unresolved double stars and bright saturated galaxies can fall in improper positions. Visual inspection of the candidate galaxies and of the bright stars is performed in order to avoid misclassification. The final catalogues are then used to produce OPTOPUS files in order to punch the plates (30 and then 50 fibers/ plate) for each OPTOPUS run. Observations and Reduction So far, observations were made in September and October 1989, March and April 1990 and September and October 1990 both for spectroscopy (3.6-m+ OPTOPUS) and CCD photometry (1.54-m Danish). CCD photometry is being performed in order to calibrate our photographic photometry. For this purpose, CCD frames have been taken for each OPTOPUS field, covering the largest possible range in magnitudes so as to minimize errors in the calibration. CCD frames are taken in both B and R colours, exposure times being typically of 15 min in B and 5 min in R. Several standard stars and secondary fields were also observed during the night. Photometric observing runs suffered partial bad weather or bad conditions and technical problems (pointing was lost several times due to computer crash) leading however to a reliable bulk of data, but on a smaller than expected number of objects. These CCD data have been processed using the IRAF package and a faint galaxy photometry package installed on SUN workstations at CFHT. The reduction yields photometric parameters (positions, ellipticities, Ras, B2s...) for several hundred of galaxies. These B and R magnitudes were then used in order to calibrate the photometric parameter P produced by ASTROSCAN. Taking into account the relative magnitude offset from plate to plate, a linear relation between Bes and log P was fitted to the combined data (goodness of fit 0.98) and shown in Figure 1. From this picture, it turns out that the dispersion in the correlation between photographic and CCD magnitudes is of about 0.25 magnitudes and that the limiting magnitude of our spectroscopic survey is about B2s=21. The OPTOPUS multifiber instrument coupled with Boller and Chivens spectrograph has been used at typical

resolutions of about 10 Å and spectral range between 3900 and 5900 Å. Since March 1990 the spectroscopic efficiency has been greatly increased thanks to a new optical configuration, an increased number of fibers (50 instead of 30) and the possibility of plugging the fibers in the plate at the desk of the observing room during the previous exposure acquisition. More recently the availability of the new Tektronik CCD detector, having a much lower read-out noise, gave another significant improvement at the system overall efficiency making OPTOPUS a real "industrial-era" z-machine. Spectroscopic data reduction, which includes fibers extraction, wavelength calibration, continuum removal and cross correlation, has been performed by means of FIGARO routines, well suited for our aim, installed in Granada and Montpellier on Vax stations. Wavelength to pixel solution is obtained on a long arc (20 mn), exposure of an He-Ne lamp taken at the beginning of each night so as to ensure a high S/N even of faint features. Relative shift is computed and applied on each subsequent arc frame. Arc frames were taken both before and after the cluster exposure to minimize errors induced by mechanical flexures of the instrument which turn out anyway to be negligible. The wavelength solution obtained by a 3rd order polynomial fit to our data leads to a typical rms per frame of 0.08 to 0.1 Å (maximum value being about 0.3, 0.4 Å). To illustrate the quality of our data, a set of wavelength calibrated spectra of decreasing quality is shown in Figure 2. The redshifts were then determined by cross-correlation technique. We found that best results are obtained using as template a galaxy spectrum belonging to our sample and having a high S/N, as can be seen by comparing Figures 3 and 4, which show the correlation peaks obtained correlating the spectra of Figure 2 with a bright galaxy belonging to our sample and with a standard velocity template (M32). The zero point can be subsequently determined by cross-correlating the galaxy template with a few well-known velocity objects of which we got spectra with different instruments. So far, about 3,000 spectra have been collected. The efficiency (number of actually determined velocities/number of spectra) has increased from about 60 % (the first runs) to about 85 % (the last ones) due to better set-up and a more efficient acquisition device leading presently to about 2,000 measured velocities. All these data have been reduced and a histogram of radial

velocity for the whole sample is shown in Figure 5. It can be seen that the data are peaked at $z=0.05$ as we may expect due to our selection criteria. The errors on the velocities are obtained following Tonry and Davis prescriptions; we get a typical value of 40 km/sec as can be seen in Figure 6 in which the histogram of the errors for all the available data is shown. It should be noted that the large scatter is due to a substantial improvement of the error value which decreased from about 60 km/sec for the first runs to about 30 km/sec due to the increased S/N of the last-run spectra. Comparison to external data is under progress. Finally, in Figure 7 we show the distribution of velocities obtained in one field of the cluster A 3662; this is a "structure" cluster, that is a cluster on which we plan to perform detailed dynamical analysis. More data are thus going to be acquired, nevertheless, even from this single field we can suspect the presence of a complex structure (two peaks?). Further data will allow a check on the reality of this feature. In conclusion, the aim of our project is to give new results both on the structure and dynamics of clusters of galaxies and on their peculiar motions with respect to the Hubble flow. For these reasons we have drawn a composite sample of more than 100 clusters, for which we plan to collect a large bulk of spectroscopic and photometric data. The Optopus multi-fiber instrument is particularly well suited to our aim. It would not have been possible to design such a large project without a large amount of granted telescope time as it is in the philosophy of the ESO Key Programmes. A Progress Report on the VLT Instrumentation Plan The VLT Instrumentation Plan The Very Large Telescope of the European Southern Observatory is the most ambitious project in the history of ground-based optical-infrared astronomy. With its four 8-m telescopes to be operated as separate units and in a combined mode and the associated array of smaller telescopes for interferometry it represents a unique technical and managerial challenge. Within the overall project, the procurement, installation and operation of a set of instruments at the different foci of the array are in itself an effort much larger than anything done in the past at ESO or at any other observatory. At the same time it is crucial to achieve the scientific goals of the project. For this reason the definition and procurement of the first-generation instruments was tackled very early in the project schedule. In June 1989, ESO elaborated and distributed widely in the community a Preliminary

Instrumentation Plan which was based on recommendations by the VLT Working Groups, set up to give advice on the scientific use of the VLT, and technical work carried out at ESO. Based on the responses and comments to this Plan, ESO prepared a revised version which was adopted by the Scientific and Technical Committee in March 1990. This Instrumentation Plan now includes ten instruments and two replicas and a tentative schedule for their implementation at the VLT. Some of the instruments are relatively well defined, for others preparatory work is under way to arrive at a complete set of specifications. A review article on the VLT instruments has been published in the Journal of Optics (1991) Vol. 22, p. 85. Excluded from this plan is the instrumentation to be designed for the VLT Interferometer. Figure 1 shows the mechanical structure of the unit telescope and the foci positions and Table 1 lists the various instruments with their assigned location. The complement of instruments at the first two telescopes can be considered as relatively frozen but the information on the last two telescopes is indicative and might be updated as the project evolves. A cornerstone of the VLT Instrumentation Plan is the participation of institutes in the ESO member countries in the construction of most of the instruments. This is a major departure from the current situation which sees the quasi-totality of the installed ESO instruments to be the result of internal development. The new approach is dictated by the need to fully involve the community in the VLT and at the same time to overcome the limitations on manpower and budget. Only about 10 % of the total VLT financial resources is reserved for instrumentation, even if it is this VLT component which at the end sets the quality and the variety of the scientific results to be obtained by the facility. To use the financial resources in an optimal way, ESO expects to pay for the instrument components to be procured from industry while the institutes will contribute with their expertise, manpower and free use of their facilities. Beside this general guideline, the mechanism of selection of institutes to be associated with the instrument projects is going to be adapted to the characteristics of the single instrument under consideration, such as the degree of definition of the specifications, the amount of expertise in that specific field existing in the community and the time available for the project. Clearly, ESO has to exercise in all cases a tight control on the quality, the costs and the development schedule

to make sure that the VLT Instrumentation Plan fully meets its scientific goals and is realized in time and within budget. In the following paragraphs, we present a short resume of the status of the different instruments and of the future actions. The VLT Instrumentation effort at ESO is organized by Jacques Beckers, Sandro D'Odorico and A. Moorwood.

2. Status of the Instrument Projects

2.1 Medium-Resolution IR Spectrometer/Imager

This is one of the two instruments in the Plan to be realized under the direct responsibility of ESO and will be the first major instrument to be installed at the VLT. It will provide for direct imaging at different scales and long slit spectroscopy over the 1-5 μm wavelength range and it is scheduled to be installed at one of the Nasmyth foci of the first unit telescope during the second half of 1996. The Infrared Instrumentation Group is currently finalizing its detailed Technical Specification and a Preliminary Design and Implementation Plan for both internal review and presentation to the Scientific and Technical Committee in November of this year. The operation modes of this instrument, to be built in two copies for the Cassegrain foci of the first and the third unit telescope, are similar to those of the two EFOSC instruments in operation at La Silla. Additional features will include a multislit unit and optics for polarimetry observations. A Call for Proposals for the Design, Construction and Installation of the two copies was distributed in May 1990. Two Proposals were received, one from a consortium of the Observatories of Toulouse, Roden and Trieste and the other from a consortium of the Observatories of Heidelberg, Munich and Gottingen. Both proposals being of excellent technical quality, at the end of the reviewing process the German consortium was selected because their offer presents a very significant financial advantage. The contract negotiations are now under way and the project work is planned to start officially in the last quarter of this year. Its schedule calls for an installation of the first copy of the instrument at the UT1 at the end of 1996.

2.3 Near-IR High-Resolution Imaging Camera

This is the first instrument to make use of the individual coude foci of the VLT and its associated adaptive optics. Its goal is to do diffraction-limited imaging and low-resolution spectroscopy in the 1-to 5-micron wavelength region. It will do that either directly at the longer wavelengths, with or without adaptive optics, or, at the shorter wavelengths, with the aid of image reconstruction algorithms developed in speckle interferometry. The Call

for Proposals for this instrument distributed in May 1990 listed only the required capabilities. It left the detailed design up to the proposers. This resulted in two very different proposed instrument realizations, one based on an all-mirror design, the other including lenses. The latter, proposed by a consortium of the Max Planck Institutes for Astronomy and Extraterrestrial Physics together with the Observatory of Turin was selected after a thorough evaluation. Its schedule calls for installation on the first VLT telescope in 1997.

2.4 UV-Visual Echelle Spectrograph

This is the other instrument to be built under direct responsibility of ESO. At least two identical copies will be built for installation at two Nasmyth foci of two unit telescopes. Based on the concept included in the VLT Instrumentation Plan, the instrument design aims at a very high efficiency and at a maximum resolving power of 40,000. Studies of the various subsystems are now under way with the goal to complete a Pre-design Report and an Implementation Plan by April 1992 for internal review and presentation to the Scientific and Technical Committee.

2.5 Multi-Fibre Area Spectrograph

The Focal Reducer/Spectrographs to be built for the Cassegrain foci of the VLT are designed to do multiobject spectroscopy over a field of 7×5 arcmin approximately. The Multi-Fibre Spectrograph will be designed to gather and guide the light of objects distributed over the larger field of view of the Nasmyth focus (30 arcminutes diameter approximately) to the slit of a stationary medium-resolution spectrograph. The area spectroscopy option of the same instrument shall be designed to provide spectroscopic data over a 2D array of points covering an area of the sky at a resolution corresponding to the best seeing conditions at the telescope. In the studies which led to the definition of the VLT Instrumentation Plan, the scientific objectives of this type of instrument were not discussed in any detail and hence the technical requirements such as e.g. the number of fibres and the optimal spectroscopic resolution are not defined. To fill this gap, ESO has distributed at the end of July 1991 a Call for Proposals for a Definition Study and a Predesign of the

Instrument	Location	Resolution	Field of View
UV-Visual Echelle Spectrograph	Nasmyth foci	40,000	7×5 arcmin
Multi-Fibre Area Spectrograph	Cassegrain foci	Medium	30 arcmin
Visible Speckle Camera	8-metre telescopes	0.01 arcsec	-

2.6 Visible Speckle Camera

To reach the highest possible, diffraction-limited resolution of the 8-metre telescopes (0.01 arcsec) it is necessary to use the speckle interferometry

techniques at visible wavelengths. These have been largely conceived and developed by scientists in the ESO member countries in the last two decades. These techniques have reached a level of maturity which warrants the implementation of a general-user facility. Although initially planned for a VLT Nasmyth focus, the Visible Speckle Camera is now planned for a coude focus since the gains in sensitivity resulting from partial wavefront correction by the infrared adaptive optics more than offsets the losses due to the extra reflections. A call for Proposals for a Definition Study of this instrument is to be released later this year.

2.7 Mid-IR Imager/Spectrometer This instrument is destined for the Cassegrain focus of unit telescope 2 where it will provide for direct imaging and spectroscopy in the 8-14 μm and possibly 20 μm atmospheric windows. Because this is a new but growing area of observational astronomy various options, particularly for the spectroscopic mode, are still being studied and a 10 μm camera with limited spectroscopic capabilities (TIMMI) is also being developed in collaboration with the Service d'Astrophysique in Saclay, France, as part of the VLT Preparatory Programme. ESO is currently planning to issue a Preliminary Enquiry early in 1992 with the aim of establishing which Institutes would be interested in participating in the development of this instrument. Depending on the actual response the intention would then be to place a Phase A study contract with an Institute or Consortium to be followed by a Development Contract based on the Phase A results and any relevant feedback from the TIMMI project obtained in the meantime.

2.8 High-Resolution Spectrographs The Instrumentation Plan presently includes three high-resolution spectrographs: an optical one to be installed at the combined focus, an infrared Multichannel FTS for a Nasmyth focus and a cryogenic infrared echelle spectrograph again for the combined focus, although it seems difficult to retain both of the latter in the basic instrument complement. Preliminary studies on possible concepts have been carried out, but both the technical data and the scientific requirements are not sufficiently defined to start action on the procurement. In February 1992 ESO is organizing a workshop focused on High-Resolution Spectroscopy with the VLT. We expect that the scientific objectives in this area of research and the different technical

concepts will be thoroughly reviewed and discussed, opening the way for a decision later in the year.

3. Upgrading the VLT Instrumentation Plan

By the end of 1992, seven instrument projects (with two of them foreseeing the manufacture of a replica) will be in various stages of development. The remaining three will be in a definition phase. The entire VLT programme will then be far advanced and detailed information will be available on cost, schedule and foreseen performance. The time will then be ripe for a critical review of the entire instrumentation plan, an assessment of the resources still available and decisions on additional instruments.

The VLT Adaptive Optics Programme

From April 24 to May 5, 1991, the current configuration of the COME-ON adaptive optics prototype system for the VLT had its last test run before it returned to Europe for a major upgrade. This 11-night run was devoted to technical tests for adaptive optics and scientific observations. The COME-ON system worked all the time fully reliably and no technical problems occurred. During the run the seeing conditions ranged from excellent to mediocre, and three nights suffered from bad weather conditions. The technical part of the programme included tests of improved and new software, observations with partial correction at visible wavelengths, and recording of wavefront sequences. Software with specialized routines depending on the available photon flux allowed an increase of servo-loop bandwidth from 10 to 25 Hz in connection with the intensified Reticon and the electron-bombarded CCD (EBCCD). With the higher bandwidth and improved modal control scheme the diffraction limit of 0.13 arcsec was reached even in the K-Band under good seeing conditions. In view of partial correction by adaptive optics for speckle and long baseline interferometry, partially corrected images in the visible wavelength range were recorded with an intensified CCD. These recordings will allow a detailed analysis of the image profiles and verification of theoretical predictions for short-and long-exposure images. For these measurements an optical path was installed parallel to the IR camera. The recorded wavefront sequences will allow to explore the spatial and temporal behaviour of the turbulence-induced atmospheric perturbations and comparisons with the theory of the turbulence. The scientific objectives of this run included the environment of young stellar objects, asteroids and planets, the search of the potential third component of Sirius A,

and luminous blue variables. From the young stellar objects, S CrA is of particular interest. It shows a clear extension. The observations of the asteroid Ceres were exciting and allowed to deduce the rotation axis of that famous small member of our planetary system. Eta Carinae was observed with a CVF and a polarizer. In the K-band, one can recognize hints of the objects observed in the visible range by speckle interferometry, which have a distance of about 0.15 arcsec from the central star. All the scientific observations require a careful data reduction and analysis. The results will be published in the near future. Immediately after the run the COME-ON system was shipped back to Europe for a major upgrade. This work will be done at Observatoire de Paris-Meudon under the direction of ONERA. The upgrades include a new deformable mirror from LASERDOT with 52 actuators, an improved wavefront sensor and EBCCD provided by LEP, a faster and more powerful control computer, and some improvements of the passive optical and mechanical components. The upgraded system is scheduled to be back at La Silla in July 1992.

A Report on the SEST Users Meeting and Workshop on Millimetre-Wave Interferometry

The second SEST Users Meeting was held at ESQ Garching on Wednesday, 22 May, the first having been held last year at Onsala. In view of the decision to develop the Paranal area for the VLT, it was felt timely to discuss the possibility of a millimetre-wave array in the southern hemisphere, and a workshop on this subject was held on the day following the Users meeting. The combination of these two meetings was obviously popular, as over 60 persons attended.

The SEST Users Meeting

The Users meeting commenced with a number of short reviews illustrating some of the recent scientific results from SEST: The Magellanic Clouds Key Programme, CO Rotation Curves of Galaxies, CO in Centaurus A, Spectral Scans of Sgr B2, Observations of Planetary Nebulae, and A CO Survey of IRAS Stars. These were followed by talks and discussion covering all aspects of the operation and performance of SEST. R. Booth summarized recent developments and future plans, and reviewed specific technical areas, including pointing, holography, receivers (heterodyne and bolometer), spectrometers, observations, calibration, and data reduction. Some of the main discussion points are summarized below.

Telescope pointing remains a matter of concern, and efforts to improve it were outlined. Inclinoimeters placed on the horizontal part of the fork above the

azimuth bearing showed a temperature dependent tilt of the telescope axis amounting to a daily variation of about 6". First attempts to improve the pointing by including the inclinometer data in the pointing solution are disappointing, giving an improvement in the rms error of only 0.7". However, these results are very preliminary. The overall pointing rms is still about 3" in both coordinates. The interferometer telescopes at IRAM show similar axis tilts, so the problem is apparently inherent in the design. The surface accuracy is another area of concern. Holographic measurements of the SEST reflector surface and its subsequent adjustment led to a significant improvement in the profile (rms 60 micron). However, a subsequent readjustment has increased the error again to around 75 micron. Unfortunately the satellite which has been used as a beacon for the holography measurements is no longer available to us, so other measurement techniques are being investigated. As for receivers, SEST continues to operate with the Schottky systems for both the 100 GHz and 200 GHz bands. The 230 GHz receiver will soon be replaced by an SIS mixer on loan from Smithsonian Astrophysical Observatory. Intermediate frequency amplifiers in both receivers have been replaced by wide (1 GHz) band HEMT amplifiers in an attempt to reduce the noise at the band edges. Another 1 GHz AOS is on order from the University of Koln. This and the purchase of a new frequency synthesizer will facilitate simultaneous observations in both the 100 and 200 GHz bands. The 350 GHz SIS receiver constructed at Onsala was completed in January and installed on the telescope for tests in April. Unfortunately it has so far proved to be impossible to fabricate SIS junctions of sufficiently small area and hence the noise temperature of the receiver is about 3000K SSB. It is hoped that better junctions will be built in collaboration with the University of Cambridge Metallurgy Department and that the receiver will be installed next year. The 1-mm bolometer receiver being built in collaboration with the Max-Planck-Institut fur Radioastronomie, Bonn, is virtually complete and will be installed on the telescope in late August. This system should then be routinely available and, because of its high sensitivity, should make more pointing sources available. Particular problems identified at the meeting were pointing jumps (possibly associated with the axis tilt), the need for better receivers at the "work horse" frequencies, and a local oscillator instability which had caused

spurious low level wings on narrow lines. The last problem has been fixed but it is important that observers are aware of it. It was suggested that a SEST Bulletin Board should be established in the Garching computer system so that users can be informed about changes and problems. Suggestions for possible future developments which were discussed included receivers at lower frequencies (<100 GHz), the 150 GHz band, and higher frequencies (410, 460 GHz) VLBI, focal plane imaging arrays, the development of a wobbling secondary, and the eventual move of SEST to the Paranal area. However, for the immediate future it was generally agreed that we should concentrate on improving the pointing and surface of the telescope, and equipping SEST with "state-of-the-art" receivers at 100 and 200 GHz.

Workshop on Millimetre Interferometry In order to provide suitable background for this subject and bring all participants up to date on relevant developments around the world, this workshop began with reviews on existing millimetre-wave interferometers: The IRAM Interferometer on Plateau de Bure, the Berkeley Interferometer, the Caltech Millimetre Interferometer, the Nobeyama Interferometer, and Millimetre VLBI. Planned and proposed developments were then summarized: the Australia Telescope at Millimetre Wavelengths, the Smithsonian Sub-Millimetre Array, plans for interferometry between the JCMT and CSO telescopes, and the NRAO Millimetre Array. The subject of a possible millimetre array near Paranal was then introduced with three talks on the site and atmospheric conditions. M. Sarazin described the site, and summarized relevant measurements made in the ESO site testing campaign. He showed in particular that the wind, an important parameter for exposed antennas working at high frequencies, appears to be lower in the valleys than on Paranal itself. A lively discussion then followed on the idea of a millimetre-wave interferometer at Paranal, introduced and stimulated by a talk by R. Booth which outlined a possible concept consisting of 10×8 m antennas arranged in an optimum configuration with regard to the equivalent interferometer beam. He considered that such an array could be built for about OM SOM and suggested that we in Europe should consider ways of raising money in order to achieve a useful millimetre array on Paranal as soon as possible. Various views were expressed by the participants. These ranged from the political through the practical to the ideal scientific concept. A case was made for

greater sensitivity, more like that of the proposed US millimetre array which would consist of 40x 80 m antennas but will cost at least 120M dollars. It was pointed out, however, that none of the existing millimetre arrays is as sensitive as that proposed by Booth and that the southern hemisphere is virgin territory for such an instrument. There was clear enthusiasm for a southern array. But how could funding be achieved? Could we approach the EEC, or could a consortium from individual countries be put together? In this context, the Japanese plans for a new millimetre array possibly in the southern hemisphere raised the interesting possibility of collaboration with the Japanese astronomers. While no clear answers were forthcoming, the idea that a working group should be set up to consider funding as well as an optimum array design was thought to be a reasonable conclusion of this first meeting on the subject. Impressions from the XXI IAU General Assembly For the first time in its 72-year history, the International Astronomical Union held its triannual General Assembly in South America. From July 23 to August 1, 1991, about 1200 astronomers from all continents met at the San Martin Centre in Buenos Aires at the River Plate. This was somewhat less than in Baltimore in 1988, but a very respectable number in view of the long travel distance to the Argentinian capital for many of the participants, and certainly sufficient to make this a most memorable and useful meeting. There was a wealth of interesting scientific talks and, as is usual at IAU General Assemblies, much discussion about policy matters of common concern. There were three very well presented Invited Discourses and seven one-day Joint Discussions, in addition to the fully packed programme of the 40 IAU Commissions. A full report will appear next year in the IAU Transactions and Highlights. However, for the benefit of those Messenger readers who were not present in Buenos Aires, we bring here a small selection of personal impressions, mostly written by participating ESO astronomers. These reports are of course very fragmentary, and it is certainly not possible to do full justice to all the exciting events at such a large meeting. Some of the accounts have been adapted from the excellent daily newspaper; the unsigned notes are by the editor.

The Inauguration The General Assembly was formally opened on July 23 by the President of the IAU, Professor Yoshihide Kozai. During the immediately preceding Inaugural Ceremony, short speeches were given, several by the organizers

and dignitaries. The Chairman of the Local Organizing Committee was happy to see so many astronomers from all over the world. He was followed by his counterpart in the National Organizing Committee, and the President of the Argentine Astronomical Association. Dr. Mendez and his collaborators carried the enormous burden of the successful organization of the General Assembly and all participants owe them a great debt, also in view of the very difficult financial situation under which the preparations were made. Following a welcome by the Secretary of Education and Culture of the Municipality of Buenos Aires, the President of the Argentinian Republic entered the hall, together with the Secretary of State for Science and Technology. After brief speeches by the Secretary of State and the IAU President, President Menem addressed the audience in his native language, of which a masterful simultaneous translation into English, the working language of Astronomy, was delivered by his interpreter. The President stressed the potential for discoveries in this very old, and yet so modern science. He emphasized the importance for the search for the deepest truths and the possibility to approach these matters from the scientific as well as from the theological point of view. He welcomed the manned exploration of space and the opening of new vistas for mankind, and he was impressed by the international character of our science. He thought that he saw "stars" in the auditorium! His speech was greeted with enthusiastic applause. The Inauguration was accompanied by a musical interlude by the excellent Brass Quartet of the La Plata University. First Planet Outside the Solar System? Perhaps the most sensational scientific news presented at the General Assembly was the announcement, by a group of astronomers at the Jodrell Bank Radio Observatory near Manchester in the U.K., of the discovery of what appears to be a planet in orbit around a pulsar. A scientific account appeared on July 25 in the scientific journal *Nature* and this news generated an enormous interest among the participants as well as the media representatives present at the GA. There were daily reports in the newspapers and TV interviews; few GA participants escaped from being asked about their personal opinions about the scientific implications of this major discovery. Observing the pulsar PSR 1829-10, discovered with the Lowell 76-m telescope in 1985, during more than three years, A. Lyne, M. Bailes and S. Shemar found that the 330-msec period is modulated with 8 msec

over a period of 184 days. Assuming a pulsar mass of 1.4 solar masses, this can be explained by the existence of a planet with 10 Earth masses in a 0.7 A.U. circular orbit around the pulsar. Most participants agreed that the observations are correct beyond any doubt, and that the above explanation is the most straightforward one. However, it seems very difficult to understand how such a planet could have survived the blast from the supernova explosion that created the pulsar somewhat more than 1 million years ago; this age is deduced from the period derivative of the pulsar. One possibility is that the planet was captured in some way. Some astronomers would not exclude that the near coincidence between the 184-day period and one half of an Earth year may have some significance and that the observed effect might possibly be explained in some other way. Whatever the outcome, this is a most exciting discovery and once again reminds us of the fact that while observational astronomy may be progressing at a breathtaking speed, nature may still have many more surprises in store for us.

A Missed Opportunity In spite of often heard claims to the contrary, IAU General Assemblies do occasionally become the stage for the announcement of important new discoveries. The most prominent case in 1991 was undoubtedly the detection of a planet around the pulsar PSR 1829-10. On July 25, it figured prominently on the first page of the daily newspaper of the General Assembly. In the articles, it was said that the presentation of the data would take place in room "M", which was not in the San Martin Cultural Center but in a building one block down the street. I arrived there in time -only to find out that Room "M" was the venue for the meeting of the Working Group on Ap Stars! In the programme part of the newspaper, I then found the correct place: back to San Martin! There, all elevators were busy, and when I finally arrived on the 4th floor, the discussion of the paper by F. Graham Smith had just begun. So, I had missed a historical event and was reminded of what I knew already: big conferences require careful preparation, also and in particular on the part of the participants. Throughout the Buenos Aires General Assembly, the participants were kept very well informed about scientific news and views, changes of programme and other practical matters by means of a daily newspaper with the appropriate name. The editors were well-known popularizers of astronomy; they were supported by a photographer. The newspaper carried

reports about the observatories in Argentina, participants' opinions about a great variety of matters -from the running of the GA to controversial scientific issues -and scientific highlights, including the possible discovery of the first planet outside the solar system and the latest news from the Hubble Space Telescope. A number of ESO-related items, from Adaptive Optics to MIDAS, were also prominently featured. To produce a four-page newspaper every day during ten days requires steely nerves and a certain ability to manage without too much sleep. In the morning of July 31, the editorial team left their office at the San Martin Centre at 2 o'clock to catch a few hours sleep at their hotel. They expected to be back at work again at 6 o'clock to continue the typesetting of the texts of IAU Resolutions which would be discussed the next day. However, when they arrived, they found the Centre on fire and it was only after some rather agitated hours that the written material and the editorial computer finally could be rescued from the office on the second floor. As a consequence, instead of the dry IAU Resolutions texts, the last issue of the newspaper carried a detailed report about this dramatic event, unique in the history of IAU.

Astronomical Technology During a meeting of Commission 9 (Instruments and Techniques), the current developments in the area of adaptive optics were reviewed. Since the last IAU GA in Baltimore, where the related technology as well as several prototype developments were discussed, adaptive optics has been successfully tested with the VLT prototype system COME-ON. The scientific results include the imaging of θ 1 Carinae and the determination of the axial inclination of the largest minor planet, Ceres. This impressive result is based on the diffraction-limited imaging with this system and was reported by F. Rigaut at a Joint Meeting of Commissions 25 and 9, dealing with the performance of infrared arrays and the scientific results obtained with these devices. The recent declassification of defense-related activities in the United States, including the successful tests of artificial guide stars, is likely to give adaptive optics in astronomy a major boost. During the Commission 9 Working Group Session on Detectors, various overviews were presented on the current status of large-format CCDs used at the major observatories. 2048x2048 pixel arrays may soon become the standard. An important question raised in connection with the use of arrays of this size, or even larger ones is: how can the current read-out time of

several minutes -during which the instrument as well as the telescope are idle -be efficiently used? In a meeting of the Working Group on High Angular Resolution Interferometry, which suffered under the circumstances after the fire on Wednesday 31, only very short status reports on ongoing projects and related technologies were given. However, this WG has a Newsletter which is jointly published with NOAO and which helps the members to stay up-to-date with the new developments.

Astronomical Education in Argentina Several meetings about Education in Astronomy were held in conjunction with the XX1st General Assembly. At the Buenos Aires Planetarium, a meeting took place on July 22 with participation of educationally oriented IAU astronomers and local teachers of natural sciences. Presentations were given about modern means and methods, by which the pupils and students can experience up-to-date astronomy and astrophysics. There was a very lively discussion and it was repeatedly stressed how important it is to stimulate enthusiasm for this extremely wide subject, by judiciously selecting and explaining those topics which are reasonably easily accessible to the students. Astronomy opens the roads to many other subjects. The relations between professionals and amateurs were discussed during a one-day meeting on July 21, also in Buenos Aires. The services of amateurs to astronomy were highly lauded and many examples of good amateur work were presented by members of an association.

Venusian Nomenclature Once a year, the IAU Working Group on Planetary System Nomenclature meets to decide about the naming of newly-discovered features on the major planets and their moons. This time, the main subject was Venus. Following the lamented death last year of the Chairman of the WGPSN, Kaare Aksnes chaired the meetings, which concentrated on the naming of the many features on the surfaces of our sister planet, recently discovered by the highly successful US spacecraft Magellan. The names must be given according to certain rules. For example, no duplication is allowed and the names must not have any political, military or contemporary religious meaning. No person can be honoured unless he or she has been dead for at least three years. Venus was the goddess of beauty in ancient Rome. It is therefore a natural decision that Venusian nomenclature shall be feminine in character, although some nasty tongues have commented on the hostile conditions on the surface. Among the names put

forward by the WGPSN and later ratified by the IAU Executive Committee were Joliot Curie, Callas, Woolf and Piaf. Less widely known are perhaps Xiao Hong, Al-Taymuriyya, Erxleben and Titibu, named after a Chinese writer, an Egyptian author, a German scholar and a Japanese Haiku poetess, respectively. A Newcomer's View Buenos Aires was my first opportunity to participate in an IAU General Assembly. The atmosphere of such a meeting, with huge numbers of astronomers working in different fields present, is very special since the discussed topics cover almost everything in contemporary astronomy. Besides the delightful Invited Discourses, my own interests were focused on extragalactic astronomy and the bordering fields. There were very informative presentations about results of space missions like ROSAT, COBE and the Hubble Space Telescope. I especially enjoyed Prof. Trumper's review of the ROSAT survey and Dr. Bohringer's talk on X-ray observations of galaxy clusters, from which it was possible to learn that the gas in these objects is often very clumpy, indicating that these galaxy clusters have not yet relaxed. Dr. Bergeron talked about the huge gaseous disks whose existence may be inferred from the absorption lines that are seen in the spectra of some quasars. They pose a puzzle to galaxy "builders", since the observed element abundances indicate a substantial enrichment when compared to the primordial abundances. Prof. Pagel told about the chemical composition of HII regions, and I found that the talks by Drs. Maeder, Fairall, Bruzual, Broufman and many others were also most inspiring. Just one month before the General Assembly, I finished a prolonged stay at the ESO Headquarters in Garching, and it was a pleasure for me again to meet on another side of the world many of my friends from this organization and also to enjoy two well-organized ESO exhibitions open during the General Assembly period in Buenos Aires. The Longest Travel Many participants in the XXIst IAU GA had long travels behind them when they arrived in Buenos Aires. However, the longest and most difficult voyage of all did not originate on another continent, but was certainly that of three young Peruvian astronomers of the San Marcos University in Lima. Leaving their native city already on July 15, they first took a bus to Tacna and then continued by car to the city Arica in northern Chile. From here they travelled to Santiago de Chile by bus and on to the Argentinian border on the Santiago-Mendoza road. To their dismay, they learned that the

high Andean pass was temporarily closed because of exceptionally heavy snowfall, so they went back to Santiago and then onwards by bus towards the south. They were lucky to cross over the mountains into Argentina at Bariloche and on they went by bus to Mendoza. Finally, after an odyssey of no less than 11 days, they arrived by train in Buenos Aires! This remarkable persistence ought to set an example for other astronomers, who may have forgotten what it means to participate for the first time in a large international meeting. The Peruvian astronomers participated in the courses given by foreign lecturers at their University within the IAU Visiting Lecturers programme which was established here in 1984, thanks to the efforts of Dr. Maria Luisa Aguilar. "The IAU General Assembly was great and it was a fantastic stimulus for us", they said on the last day, "it was definitely worth the journey!" However, they were slightly worried about the return trip to Lima. About Cephei I had the privilege of reporting to Commission 27 (Variable Stars) about the work of two young Polish students, A. Pigulski and D. Boratyn, both at the Wroclaw University. B Cephei is the prototype of the class of Cep variables. It has a pulsation period of 0.1905 days. It is also a member of a multiple system; there is one visual companion at magnitude 13.4 and another one with separation decreasing from 0.25 to 0.07 arc-seconds in less than 20 years (as seen by speckle interferometry). The pulsation period of Cep underwent a sudden decrease of about 0.00001 day around 1920. Period changes are common in Cep stars, but could never be satisfactorily explained by any mechanism. However, Pigulski and Boratyn have now analyzed all photometric and spectroscopic data which have been collected during more than 70 years. They were able to show that the period variation of Cep can be completely explained by the light-time effect induced by the motion in the binary system. From this, they also obtained a preliminary solution for the orbit of the system, which has a period of 92 years. This work not only represents a breakthrough in this field, it also proves that it is worthwhile to observe bright stars, and that it is crucial that such observations continue during many decades. Only small telescopes are needed for this type of work. White Dwarf Masses At one of the Commission meetings, Dave Latham reported on the work of his Argentinian student Guillermo Torres who had analyzed the radial velocity curves of several dozen

metal-poor halo G-type dwarfs. The purpose was to study the effect of binarity on the astrophysically very important Initial Mass Function (IMF), i.e. the distribution of masses of newborn stars. No difference was found between single field stars and the companions to the sample stars. The only disturbing feature was a narrow peak at about 0.6 solar masses for the companions! After a short confusion, however, the answer was quickly found: having originally been the more massive components, these stars had already evolved to white dwarfs which typically have a mass of 0.6 solar masses. Several observational tests have meanwhile confirmed this hypothesis, and the IMF is therefore not in jeopardy.

Strange Sounds in the Sky The participants in a meeting of Commission 22 (Meteors and Interplanetary Dust) learned that a very elusive phenomenon, by many thought to have a non-physical origin, after all is a real physical effect. The President of this Commission reported how he finally, after more than a decade of research, has been able to show that reported sounds, allegedly heard in connection with passing bright fireballs, are not just a psychological effect, but can be explained by the little-known "electrophonic" phenomenon. Chinese observers described hearing such sounds already in the year 817 A. D. More recently, in 1978, a strange sound (a "swish") was heard by as many as one third of the witnesses when a fireball of magnitude -16 passed over an area of Sydney in Australia. However, the meteor moved high above the denser parts of the atmosphere through a near-vacuum, so how could sound waves be generated? Electromagnetic radiation from large meteors had been searched for without success, but the President decided that it would be worth to investigate this possibility further. A careful examination of the radio spectrum revealed that radio emission at frequencies in the audible range was not ruled out. Indeed, tests in "quiet" (anechoic) chambers with human subjects exposed to an electric field varying at audio frequencies revealed that some had a far lower auditory threshold than others for perceiving such sounds; it was noted that floppy or frizzy hair was a great help. The explanation is that hair (and some other materials) may sometimes act as transducers, converting the radio energy directly into perceptible sound. A definite proof that this explanation was correct was finally obtained when a Very-Low-Frequency signal was actually recorded in Japan at the exact time of a bright Perseid fireball. It is believed that

this "electrophonic" phenomenon is also responsible for reports about sounds perceived at the time of especially intense auroral displays. The era of Audioastronomy has opened!

Molecular Hydrogen in Galaxies

During the scientific sessions of the Working Group on Dynamics and Internal Motions in Galaxies (Commission 28), R. Allen summarized the recent results from high-resolution radio observations of spiral galaxies relating to density waves. The new radio maps of Messier 83 and 51 make it possible to compare the position of different gas components relative to the dust lanes in detail. This suggests that H I may not be the most important gas component, but is rather a product of molecular H in which star formation occurs after a shock. Furthermore, high-resolution polarization measurements indicate the direction of the magnetic field which can be used to determine the gas flow.

Minor Planet 5000 Named for IAU

The naming of minor planets is the responsibility of Commission 20 (Positions and Motions of Minor Planets, Comets and Satellites). At the time of the General Assembly, 4877 minor planets in the solar system had been numbered and as the 5000th discovery was approaching, this Commission arranged among its members a competition to decide upon the name of this object. The ballots were opened and counted on July 24, and the name "IAU" was found to be the winner, by a small margin. Runners-up were Hipparch, Ptolemaeus and Pascal. It is expected that the number 5000 will be assigned to an object by the Minor Planet Centre next November.

Dangerous Impacts

Recent studies have shown that there may be many more minor planets in earth-crossing orbits than previously thought. Improved detection techniques have resulted in an ever increasing rate of discoveries of such objects. This has led to an upward revision of the estimates of the rate of collisions with the Earth. It is also now a widespread belief that the catastrophic event which contributed to the extinction of the dinosaurs 65 million years ago was probably caused by the impact of a minor planet or comet with a diameter of perhaps 10 kilometres. Smaller impacts are also known, like the 10 Megaton Tunguska event in 1908. The greatest risk to the Earth's population is found to be associated with impacts of minor planets in the diameter range from 0.5 to 5 kilometres. There may be more than 10,000 such objects in orbits which from time to time bring them very near the Earth. We were reminded of this when a 10-20-metre "boulder" passed within

200,000 km of the Earth in early 1991. In view of the mounting interest in this area, as expressed by solar-system astronomers from several countries, IAU Commissions decided to request the creation of an IAU Working Group on Natural Near-Earth Objects (WGNEO). A corresponding resolution was formulated and was passed by the General Assembly. The WG will endeavour to stimulate cooperation among the various groups of observers and theoreticians and will act as an international focus point for the world-wide efforts to learn more about this type of celestial objects. The Chairman of the WG is Andrea Carusi, who is also President of Commission 20 (1991-94).

The Molecular Mass in the Galaxy

Among the many interesting items discussed by Commission 34 (Interstellar Matter) were the conditions prevalent in the IS Medium that lead to star formation and, on the other side, the interactions of newly formed stars with their immediate neighbourhood. A very central question to this field is: how large is the mass of molecular material in the Galaxy? A consensus is still lacking about the value of the conversion factor required to obtain the molecular hydrogen mass from CO observations. It may span an order of magnitude, depending on the metallicity of the material. If Contact, then ...

During a meeting of IAU Commission 51 (Bioastronomy), a "Declaration of Principles Concerning Activities Following the Detection of Extra-Terrestrial Intelligence", accepted by the International Academy of Astronautics, was discussed. Although this event may still seem very unlikely and extremely remote to many, it does no harm to be prepared. The text is therefore reprinted here in an abbreviated form, in particular since it cannot be excluded that one of our scientist-readers may need to consult it sometime in the future. The Declaration begins with the recognition that the search for extraterrestrial intelligence (SETI) is "an integral part of space exploration and is being undertaken for peaceful purposes and for the common interest of all mankind". Since any initial detection may be incomplete or ambiguous, "it is essential to maintain the highest standards of scientific responsibility and credibility", and various principles of behaviour are laid down in what follows:

Any individual or institute believing that any sign of extraterrestrial intelligence has been detected, should seek verification and confirmation before taking further action. Before making any such announcement, the discoverer should promptly notify all other observers or organizations which

are parties to this Declaration. No public announcement should be made until the credibility of the report has been established. The discoverer should then inform his national authorities. After concluding that the discovery is credible, the discoverer should inform the Central Bureau for Astronomical Telegrams of the International Astronomical Union, and also the Secretary-General of the United Nations. Other organizations to be notified should include the Institute of Space Law, the International Telecommunication Union (ITU), and Commission 51 of the International Astronomical Union. A confirmed detection of extraterrestrial intelligence should be disseminated promptly, openly and widely through the mass media. All data necessary for confirmation of detection should be made available for further analysis. All data relating to the discovery should be recorded, and stored permanently in a form which will make them available for further analysis. If the evidence of detection is in the form of electromagnetic signals, the parties to this Declaration should seek international agreement to protect the appropriate frequencies. Immediate notice should be sent to the Secretary-General of the ITU in Geneva. No response to a signal or other evidence of extraterrestrial intelligence should be sent until appropriate international consultations have taken place. The SETI Committee of the International Academy of Astronautics, in coordination with Commission 51 of the IAU, will conduct a continuing review of all procedures relating to the detection of extraterrestrial intelligence and the subsequent handling of the data.

On the Future of Existing Photometric Systems Astronomical photometry is facing a crisis: every new detector has caused a proliferation of new photometric systems, lacking proper standardization and calibration. From past experience one can only fear that the situation will become worse with the growing application of CCD's to astronomical photometry. The time had therefore again come for Commission 25 (Stellar Photometry and Polarimetry) to re-examine the fundamentals of photometry. The current situation can only be cured if we: return to the teaching of the fundamentals of photometry; design and develop standardized and user-friendly reduction programmes, incorporated in large data-reduction packages; propose a new standard system that is open and well-sampled, and which allows transformation back into well-known systems such as UBV, or Geneva; have many (500) filter sets manufactured and

distributed; have small networks of automatic telescopes to set up the system; design a New Technology Photometer (Fourier Transform Spectrometer, Fabry-Perot, etc.) for use at large (or very large) telescopes, and push for the realization of a standard photometric light source in orbit. An International Observatory in Antarctica? Astronomers have a tradition of placing observatories in remote places. At this General Assembly, a Joint Commission Meeting was held about the "Development of Antarctic Astronomy". Fifteen speakers from ten countries surveyed the possibilities of establishing an international astronomical observatory in one of the most hostile areas on this planet, on the high plateau in the eastern part in Antarctica, more than 4000 metres above sea level. Here the temperature may drop below -80°C , the humidity is extremely low and the mean wind speed is only about 3 m/sec. Such a site would offer several advantages, especially for work in the infrared part of the spectrum and also at millimetre wavelengths. The continuous antarctic night would allow 24-hour access to a large part of the southern sky and the seeing may possibly be very good. Reports were presented about astronomical research which has been carried out at the South Pole with smaller, transportable telescopes. A Center for Astrophysical Research in the Antarctic (CARA) has recently been set up in the U.S.A. The meeting decided to submit a formal resolution to the second session of the General Assembly about the creation of an IAU Working Group to study all aspects of antarctic astronomy and to work towards the establishment of an international observatory on that continent. The resolution was duly passed by the General Assembly on August 1.

ESO-MIDAS at the General Assembly The ESO exhibition at the General Assembly of the IAU was enhanced with a portable laptop computer, running under the UNIX operating system and with the X11 windowing software. At regular intervals, this equipment was used to demonstrate ESO's image-processing system: ESO-MIDAS. In addition to these demonstrations some samples of the MIDAS documentation were displayed. During the GA many people showed their interest in the MIDAS project, in particular astronomers from developing countries. Among these, roughly 25 people made requests to receive detailed documentation about MIDAS. Also, in the GA newsletter, an article was published that gave an overview of the

functionalities at the command level; it also described the existence of the MIDAS programming environment. Given the very positive experiences obtained from the MIDAS demonstrations, plans are now being developed for a more advanced display of information about MIDAS. For this purpose, a set of exhibition panels will be produced which will, whenever appropriate, be included together with the laptop in the already existing ESO exhibition.

New Working Group on Wide-Field Imaging The interest in photography among astronomers has been steadily decreasing during the past years. This is of course due to the advent of new and more sensitive detectors, like CCDs. The IAU Working Group on Photography has recognized this development for some time and has discussed what to do, also in view of the continued need to ensure the availability of large photographic plates for the world's big Schmidt telescopes. At the same time, there has been a growing awareness of the usefulness of digitized, computer-readable sky surveys. While they may not retain the full information of the corresponding photographic plates, they are certainly very handy for the convenient production of finding charts, etc. Commission 9 therefore decided to discontinue the WG on Photography, and to create a new Working Group on Wide-field Imaging with the following main areas of concern: 1. Sky Surveys and Patrols; 2. Photographic Techniques; 3. Digitization Techniques; 4. Archival and Retrieval of Wide-field Data. As can be seen, the new WG incorporates the functions of the earlier WG on Photography, but places the use of photography within a wider context. The Chairman of the IAU WG on Wide-field Imaging is R.M. West. One of the most urgent tasks of the new WG will be to look into the possibility of re-activating regular patrols of the northern and southern sky; such patrols were until recently carried out at Harvard and Sonneberg and have repeatedly shown their value for the study of the long-term behavior of variable objects. This will include a careful consideration of the desirable characteristics of such patrols, for instance field size, angular resolution, limiting magnitude, spectral passband(s), observing strategy and, of course, the corresponding choice of detector and telescope optics.

Fire in the Martin Centre On the next-to-last day of the General Assembly, the San Martin Centre caught fire. When the participants arrived for the morning's sessions, they were greeted by black clouds of smoke and hectic

activity by the municipal fire brigade. It later became known that the cause most probably was a cigarette butt in a trashcan in the subterranean parking lot, right under the exhibition area; the floor under the ESO Exhibition got so hot that parts of it began to bulge. Fortunately, there were no casualties, although two of the brave firemen had to be treated for smoke poisoning. It was of course excluded to hold the scheduled meetings in that building, and it looked as if the rest of the GA program was in jeopardy. However, the local organizers, together with the IAU General Secretary and his secretarial staff, did a most wonderful piece of improvisation and were able to secure rooms in a neighboring Plaza, so that only a handful of meetings had to be canceled. In particular, the important Joint Discussion on the results from the Hubble Space Telescope could start with a delay of only 90 minutes and by running on a continuous schedule until 7 o'clock that evening, virtually all of the presentations were saved. Space Astronomy "A crisis in space astrophysics" may be too extreme a view, but accurate atomic data are lacking for many ions and transitions, now observable with high spectral resolution, particularly in the UV with the Hubble Space Telescope. A Joint Discussion on Needs and Availability of such data brought together once again the group of physicists whose forefathers founded the Astrophysical Journal in a similar situation. A whole day of mostly science presentations on "First Results from HST" was to conclude the series of Joint Discussions. Reportedly, the project with its "hot" topics was, however, not responsible for the unfortunate breakout of fire early that Wednesday morning in the Conference Centre. New Results from the HST The last day of the IAU General Assembly was for me the most memorable, not only for the fire, but especially for the HST presentations in Joint Discussion VII. The auditorium was better than the standard ones at the St Martin Centre; the presentations were more professional: you could easily hear and see what was being presented, people had thought about the structure of their talks as well as about the sequence of the several related topics. The UV spectroscopy from the GHRS, which I saw there for the first time, was superb, a dream come true for many an astrophysicist. The few FOS results Colin Norman showed in his summary also were most promising. The FOC is clearly a well-designed instrument and, given enough telescope and computer time, moves frontiers. Contrarily,

in my view, the huge effort going into producing "beautiful images" is of questionable value. The HST PR people still use what I call the "Weight Watchers' trick of before and after photos" which can be very deceiving, if not downright dishonest. For example, two images of the center of the globular cluster 47 Tucanae, from the 2.2-m at La Silla and from the HST, which were shown on the front page of issue No. 10 of the daily IAU paper are not at all comparable. If we had spent a few more minutes of 2.2-m time, gone to the trouble of determining the PSF (image shape) of that exposure and made, say, 20 iterations with the Lucy image-sharpening routine, our picture would have been very similar to that of the HST, at least the "contrast" between the two would have been of a different order altogether. My thoughts that Wednesday, in addition to the pleasure of these results, were of the fantastic telescope HST would have been if its primary mirror had been up to the original specifications. COSTAR may still do it, after 1993. It is also an incentive for us at ESO to continue to perfect the NTT and, not the least, to build the VLT to specifications and according to the established schedule. We must also strive to learn how to make the best possible use of the superb seeing on La Silla and on Paranal whenever it occurs. The HST is a complement to our telescopes; it is also a challenge to the ESO team's ingenuity.

New IAU Members

The General Assembly admitted about 800 new individual members, bringing the total to about 7,500. It is predicted that the IAU may have 10,000 members by the turn of the century, causing some concern about the practicality of retaining individual membership, a most valuable IAU feature, and unique among the international scientific unions.

Resolutions

The IAU General Assembly passed 7 resolutions of which the full texts will duly appear in the IAU Bulletin. Several of these are concerned with adverse influences on observational astronomy, in particular the continuing fight to avoid the intrusion of orbital and ground-based radio transmissions into the astronomically important wavelength bands, e.g. the OH windows. One of the resolutions is extremely complex and deals with the new Reference Systems which will now be adopted in astronomy. For the first time, they will specifically incorporate relativistic effects. In practical terms, they also define the transition to the J2000 coordinate system in astrometry.

New IAU Executive Committee

Following the formal election procedures during the second

GA session on August 1, the new Executive Committee now consists of: President A. A. Boyarchuk; President-elect L. Woltjer; Vice-presidents D.S. Mathewson, F. Pacini, V. Radhakrishnan, M. Roberts, Ye Shu-hua, J. Smak; General Secretary J. Bergeron; Assistant General Secretary I. Appenzeller; Advisors Y. Kozai and D. McNally. L. Woltjer is former Director General of ESO; Franco Pacini is currently President of the ESO Council, and also Imme Appenzeller has close connections to our organization as a member of various committees.

Format of the IAU General Assemblies

The question of the format of future IAU General Assemblies has surfaced again and again during the past years. Most IAU members would agree that there is room for improvement, but many disagree about how this can best be accomplished. In the early days of the IAU, the General Assemblies were very central events in professional astronomy. Before the advent of cheap and fast intercontinental travel, they were the foremost occasions for meetings among astronomers from different countries. In view of the relatively small number of participants, a few hundred at most, they were perfect frames for intensive consultations about scientific progress and international collaboration. These days are long past, and many astronomers now think that their sparse travel money is better spent on participating in specialized Symposia and Colloquia than on the IAU General Assemblies which they feel are scientifically less interesting and moreover heavily burdened with administrative procedures for which they have little sympathy. This is particularly true for some younger astronomers, and the General Assemblies are in danger of losing one of their most important earlier functions, that of introducing newcomers to their more senior colleagues from other parts of the world and tying personal bonds among the world's astronomers. Some astronomers also think that General Assemblies and the associated, immediately preceding or following specialized Symposia and Colloquia simply last too long - who can afford to be away from their home institute during four full weeks? The result is obvious: the IAU General Assemblies are attracting relatively fewer and fewer participants. From perhaps 75% of all IAU members in the early days, no more than 15-30% of the members were present at the Assemblies after 1976. So what to do? In view of the recognized need for certain administrative functions of the IAU, as the adoption of important resolutions on

matters that concern astronomy worldwide, any shortening would undoubtedly mean even less time for science and therefore also less participants. However, the new IAU General Secretary, Jacqueline Bergeron, in her speech of accession during the second session of the General Assembly on August 1, said that she would attempt to increase the scientific content of the next General Assembly in 1994, by having several three-day Symposia on particularly relevant topics within the GA framework. If such a change would be agreed on, no other IAU meetings would be held within several months of the General Assembly, in the hope that this would ease the participation for more members. Her ideas were welcomed by many IAU members, but others expressed some reservation, fearing that some astronomers may decide to participate in one of these GA Symposia only, and not in the entire GA, as is the case in some other international scientific unions. Whatever the outcome, future IAU General Assemblies are likely to be different.

How Hot are the Molecular Clouds at the Galactic Centre?

Introduction

The refinement of infrared and radio technology and the new ground-based facilities have allowed to penetrate the interstellar gas which hides the Galactic Centre in the UV and optical light. These observations have revealed the complex nature of the central parts of our Galaxy and have raised some puzzling questions. In particular, the heating mechanisms of the Giant Molecular Clouds are not understood; the molecular hydrogen density is high (although not so high as in the hot cores of galactic clouds) and it seems that a pervading high temperature could be found irrespective of the galactocentric distance. Many estimates of these temperatures have been attempted; for example, Wilson found a rotational temperature of metastable levels of NH_3 as high as 175 K (and this is an underestimate of the kinetic temperature); Morris found uniformly high temperatures 30-60 K over a few hundred parsecs around the Galactic Centre. A high level of the kinetic temperature is confirmed by the detection of emission lines of SiO as seen in our previous observations. The molecule SiO has been searched in several galactic molecular clouds; it has been observed only toward sources with a kinetic temperature greater than 30 K. Here, we report on recent observations of the 20 and 50 km/s clouds and we shall focus on the determination of the temperature through a large part of the clouds.

What We Observe

We used the

15-m radio telescope SEST operated conjointly by ESO and Sweden. The observations were done in March 1991 with good weather conditions. The 3-mm receiver system consists of two cooled Schottky mixers covering the frequency band 80-115 GHz with an SSB receiver temperature of about 300 K; the image band is attenuated by more than 10dB. The 1-mm receiver is built in the same way with a receiver temperature of about 700 K. The observer tunes the receiver from the control room with a friendly tuning programme in less than ten minutes in most cases. The spectra are analysed simultaneously by two acousto-optic spectrometers of high (0.04 MHz) and low (0.7 MHz) spectral resolution. The first one is of limited spectral range (80 MHz) and is useful to study the line profiles while the second one has a larger bandwidth (500 MHz). All results are reported in T^* , the antenna temperature outside the atmosphere. The system temperature varied between 350 and 700 K at 3 mm, with a corresponding noise level of 50 mK in 4 minutes of integration. The sky subtraction was done by switching the telescope between the source and an adjacent position in the sky. First, we moved the telescope to a position known to be free of emission: since this procedure gave bad baselines and a high noise level in some spectra, we preferred to use a rotating mirror to switch the beam to a position 12' away at a rate of 6 Hz. This procedure gives good baselines but may alter the strong lines if there is some emission in the reference beam. We observed some lines both in the SgrA and SgrB2 clouds to compare with existing line surveys of the SgrB2 cloud and to check the temperature scale. We found general good agreement, except between the line intensities towards our extended sources measured with the SEST and with the IRAM 30-m telescope. The half-power beam size of the SEST varies from 44" (115 GHz) to 65" (80 GHz), so we chose a map spacing of 40". The observations were centred on the direction α, δ (1950) = 17h 42m 29.4s, -29°03'31" for the 20 km/s cloud and α, δ (1950) = 17h 42m 40s, -28°58'20" for the 50 km/s cloud. The total areas covered in the SiO map are respectively 160"x 160" and 120"x 120"; assuming a galactic centre distance of 8.5 kpc, these areas are roughly speaking 7 x 7 pc for the 20 km/s and 5x5 pc for the 50 km/s. In Figure 1, we present the spectra obtained toward each offset for the two clouds. The spectral range (87.1 to 86.7 GHz for the rest frequency) is such that the $J=1-0$ transition of HN^13C (87.091 GHz), the $J=2-1$ transition of SiO (86.743 GHz) and the

$J=1-0$ transition of H^{13}CO^+ (86.754 GHz) are selected. On both maps, we can see that SiO is present everywhere and is fairly intense (except in the last offset of Sag A 50 km/s which was not observed); these galactic centre molecular clouds are the unique sources of widespread SiO in our Galaxy. We selected 3 positions in each cloud to observe more molecular lines. As we are interested in the temperature, some examples of the spectra we use for our purpose are presented. Figure 2 gives the $J_k=6-5$ ($K=0-3$) and the $J_k=5-4$ ($K=0-3$) transitions of CH_3CCH and the $J_k=5-4$ ($K=0-4$) of CHON for the 20 km/s; these symmetrical top molecules were chosen because they are supposed to be good thermometers for the Interstellar clouds. In each case, only the lines for $K=0, 1$ and 2 are clearly identified; for $K \geq 3$, the signal-to-noise ratio is bad and we have to take the numerical values that we derive with a pinch of salt. Figure 3 gives the same transitions (with same remarks) for the 50 km/s. What We Deduce and could give an estimate of the kinetic temperature. Table 1 and Table 2 summarize the results for the 3 offsets in each cloud and also give the average value. One can also find the total column density of each molecule; it has been obtained by using our observations and the partition function computed for the temperature we determined. The magnitude of the CH_3CN column densities of the 20 km/s cloud are of the same order as those obtained by Turner for Sag B2; for the 50 km/s cloud, the column densities are somewhat higher. In both clouds, the methyl acetylene column densities are of the same order as those found by Churchwell and Hollis in Sag B2. Some remarks must be made. CH_3CCH seems more thermalized than CH_3CN , which can be understood if we recall that the electric dipole moment (and as a consequence the probabilities of radiative transitions) of CH_3CN is larger (3.9 D) than that of CH_3CCH (0.78 D): in this last molecule, the level populations are mainly governed by collisions. This is compatible with a numerical molecular hydrogen density $\sim 10 \text{ cm}^{-3}$. Concerning the galactic clouds themselves, the Tables 1 and 2 show that the kinetic temperature must definitely be high in these clouds, $\sim 25-70 \text{ K}$ in the 20 km/s, if we take the average value, $\sim 80-90 \text{ K}$ in the 50 km/s cloud. On the other hand, this "high" temperature is pervading in a fairly extended part of the clouds (at least 5 pc), which put some constraints on the heating processes. The Absolute Magnitude of RR Lyrae Stars RR Lyrae stars

are one of the standard indicators in the distance scale. Knowing their absolute magnitude, we may determine such basic astronomical parameters as the distance to the Galactic Centre and the distances and ages of the Globular Clusters, which have a cosmological connection with the Hubble constant. They have also been observed in other galaxies of the Local Group where they serve as an important check on other distance indicators, notably Cepheids. Despite intensive observational efforts over many years, there is still uncertainty in the absolute magnitudes of these stars, and in particular over the variation of the absolute magnitude with chemical composition. RR Lyraes have $[Fe/H]$ in the range from solar to several hundred times deficient, and some work suggests there is no variation of absolute magnitude with $[Fe/H]$, whilst other work suggests the variation may be as much as one magnitude. Among the methods used to determine the distance and the absolute magnitude of RR Lyrae stars, the Baade-Wesselink (B-W) method is particularly powerful since it is the only one that uses the intrinsic properties of the star, i.e. luminosity, colour, and radial velocity variations over the pulsation cycle. In its original formulation the B-W method uses the light and colour curves of the variable to derive the variation of the angular diameter over the pulsation cycle. The radial velocity curve is used to derive the variation of the linear radius. From the comparison of the variations the distance and the absolute magnitude of the star are then derived. In 1986 two of us started at ESO an observing programme aimed at the collection of visual photoelectric photometry and radial velocity data to apply the Baade-Wesselink method to a number of field RR Lyrae stars. In collaboration with Dr. Prevot of the Observatoire de Marseille and Dr. H. Lindgren of ESO, field RR Lyraes have been observed with the 1-m ESO telescope and the 1.5-m Danish telescope equipped with CORAVEL. The data for these stars and the B-W analysis of some of them have been published. Recently it has become evident that the use of infrared JHK photometry allows a more accurate application of the B-W method. As shown by the use of the infrared magnitudes and colours (in particular K and V-K) minimizes the discrepancy between photometric and spectroscopic radius determinations on the descending branch of the light curve, which is presumably due to the presence of non-LTE conditions in the pulsating atmosphere. This effect inhibits the application of the B-W method in the visual

region over that phase range, but does not significantly affect the infrared region which falls in the Rayleigh-Jeans tail of the energy distribution and is therefore less sensitive to the temperature and its "anomalies". We have therefore taken JHK observations for three of the previous objects, namely RV Phe, W Tue and UU Cet, and their absolute magnitudes have been determined using two slightly different formulations of the B-W method, i.e. (a) the Surface-Brightness (SB) method, and (b) the Infrared Flux (IF) method. These are at present the two most accurate and widely used techniques to derive $M_V(RR)$, and both are capable of determining the absolute magnitude with an error of ± 0.15 - 0.20 mag. Although this degree of accuracy is not sufficient to resolve some of the important questions related to the variation in absolute magnitude of RR Lyrae stars with period and metallicity (such as the age of globular clusters with a smaller uncertainty than the current 2-3 Gyrs), it can be substantially improved by enlarging the statistics, i.e. applying the methods to as many stars as possible.

Observations Both SB and IF methods require very accurate photometric (BVRIJHK) and radial velocity data as input parameters, the knowledge of reddening and metallicity of the star, and a model atmosphere for the proper metallicity and gravity. The infrared light curves for the three stars were collected between October 19 and 28, 1990 using the 1.0-m telescope at the European Southern Observatory. The BVRI photoelectric photometry and the radial velocities of the three stars had been obtained in previous observing runs at La Silla. The visual photometry is on the Johnson-Cousins photometric system. The Walraven photometry used in the IF method is by Lub. The radial velocities were obtained with the CORAVEL photoelectric scanner. The accuracy of the CORAVEL radial velocities of an RR Lyrae star depends on the metallicity, the average colour of the star, and the phase at which v is measured, the correlation dip being shallower and less well defined when the star has earlier spectral type. Typical accuracies for individual data points are ± 1 - 2 km/sec for the radial velocity data, and ± 0.01 - 0.02 mag for magnitudes and colours. The photometric and radial velocity data for one of the stars, namely W Tue, are shown in Figures 1 and 2 as an example.

Analysis and Results For the three programme stars, RV Phe, W Tue and UU Cet, we have adopted respectively $[Fe/H] = -1.5$, -1.35 , and -0.90 , and $E(B-V) = 0.015$, 0.005 and 0.025 . We have then

applied the IF and SB methods as described in detail, avoiding the phase interval around maximum light since the stellar atmospheres are presumably in non-LTE and static model atmospheres are therefore not adequate to describe their characteristics in that interval of phase. During the ascending branch of the light curve there is considerable evidence for a shock wave in the upper atmosphere of RR Lyrae stars, as indicated by emission in the Balmer lines and an ultraviolet (U-B) excess. This is clearly shown in Figure 3 where we report three orders from the CASPEC echelle spectrum of an RR Lyrae star in the globular cluster M4 that one of us had obtained in a previous observing run at La Silla. The spectrum corresponds to $q=0.96$, i.e just before the star reaches the maximum light. The presence of these emissions is generally interpreted as due to shock waves caused by the pulsation being "out of phase" in the higher layers of the atmosphere. As material from the new pulsation cycle drives outwards it collides with material falling in from the old pulsation cycle. The resulting collisions produce both the Balmer line emission and the (U-B) excess. Figure 3 shows that some emission occurs in weak metal lines as well. To avoid these problems, the IF method has been applied over the phase interval $\Delta q=0.05-0.85$, the SB method plus V-K colours has been applied over the phase interval $\Delta q=0.10-0.80$, and the SB method plus visual colours (i.e. V-R and V-I) has been applied on an even more restricted phase range (varying from star to star) to avoid additional distortions in the flux distribution that do not normally occur in the infrared range. The fits for W Tuc are shown in Figure 4 as an example. The angular diameters have been derived using as temperature indicators the V-R, V-I and V-K colours respectively in the SB analysis, while the entire spectral range has been used in the IF analysis. The use of the above relationship to estimate globular cluster ages leads to 16 and 19 Gyrs for metal-rich and metal-poor clusters respectively on the assumption that $[CNO/Fe]=\text{solar}$, and to slightly lower values depending on the amount and degree of metallicity dependence of O enhancement, if any. The obvious further step with respect to the work done so far on field stars is to extend this type of analysis to globular cluster RR Lyraes. Since in a given cluster RR Lyraes are all at the same distance and have the same metal abundance, the study of a sufficiently large number of them can provide a more accurate determination

of the average absolute magnitude in each cluster and of its dependence on metallicity. Globular cluster variables are however much fainter than their field counterparts, and the observations are correspondingly more difficult and time-consuming. In particular CORAVEL, in its present configuration, cannot be used for RR Lyraes fainter than $V = 12.5\text{--}13$ mag, while the brightest RR Lyraes in globular clusters at minimum light are $V13.5$. For this reason very few results have appeared on this topic, although considerable effort is being devoted to it. We have started a programme on the RR Lyrae variables in M4, and have collected BVRIJHK photometry and CASPEC high-resolution spectra for 4 variables, covering the entire pulsation cycles. The data are presently being analysed and will be the subject of a forthcoming paper.

Galaxy Photometry with SEST: How Big Are Galaxies at Millimetric Wavelengths?

Introduction

The information currently available on the electromagnetic emission of normal galaxies at long wavelengths is still quite sparse and serious discrepancies are found among different observations. Only very recently the improvement in instrument sensitivity has allowed exploration of the galaxy submm-mm continuum. Data on millimetre continuum emission of galaxies are mainly confined to active galaxies (both AGNs and starburst galaxies), because of their enhanced nuclear emission, while only a handful of normal spirals have been observed so far at these wavelengths. The IRAS surveys have shown that the far-IR spectra of late-type galaxies are mainly of thermal origin, due to dust grains present in their interstellar medium. However, far-IR measurements are not enough by themselves to estimate the overall dust content, since an important fraction (often more than 50 %) of dust is expected to be colder than 20 K and therefore its emission lies at approximately 100 μm . Therefore, in order to study the characteristics of cold dust, which are important to understand both galactic evolution and star-formation processes, submm-mm photometry is needed. The continuum flux in this spectral range is linearly related to the temperature, mass and opacities of dust grains, and its measurement provides an alternative way to estimate these parameters. The exploitation of the high sensitivity achievable by bolometer detectors, together with the high spatial resolution of large antennas such as SEST (with a FWHM of $24''$), is expected to strongly improve our knowledge of the galaxy spectra and of the total amount of dust present in the interstellar medium, its

spatial distribution within the galaxies and its relationship with other basic components, such as the atomic and molecular gas, the stars, etc. We have started an observational programme whose aim is the investigation of the 1.2-mm continuum emission of a complete sample of galaxies selected from the IRAS Point Source Catalogue, for which optical photometry and spectroscopy are partially available. The sample comprises 61 galaxies with a 60-mim flux above 2 Jy in the sky region delimited by the equatorial coordinates $21^{\text{h}} < \text{R.A.} < 5^{\text{h}}$ and $-22.5^{\circ} < \delta < -32^{\circ}$. The completeness of the sample ensures that unbiased estimates of the crucial parameters (such as the average mm to far-IR wavelength flux ratio and the bivariate luminosity distributions) could be obtained. This will eventually allow the determination of the local luminosity functions of galaxies at $\lambda = 1 \text{ mm}$. Several important applications of this analysis can be envisaged. Let us mention two among others. (1) A reliable determination of the galaxy local luminosity at $\lambda = 1 \text{ mm}$, added to observations of the extragalactic background being currently performed by the COBE satellite, and to ground-based millimetric surveys, planned in the near future with the use of bolometer arrays, may tell us something new and fundamental about the cosmic evolution of galaxies and of their dust content. (2) A better knowledge of galaxy continuous spectra in this energy domain will allow to refine the estimate of the contribution of known discrete sources to the cell-to-cell fluctuations of the Cosmic Microwave Background (CMB) at small and intermediate angular scales. A successful observational run performed last year at SEST has already provided us with some reliable detections and upper limits for half of the sample objects falling in the δ range from -22.5 and -26.5 . We briefly report here on results of these observations and on problems raised by a comparison with previously published data. Observations with the SEST Telescope The observations have been performed during September 1990 using the 15-m SEST telescope at La Silla equipped with the He bolometer of the MPIR (Max-Planck-Institut für Radioastronomie). The filter set coupled to the atmospheric transmission window provides an effective wavelength around 1.25 mm. The beam size is $24''$ (HPBW). Source position was found by pointing a nearby radio-loud quasar with strong millimetric fluxes. Pointing accuracy was most of the time better than $2''$ and was checked each half an hour. Beam-switching is

achieved by a chopper wheel located in the receiver cabin, switching the beam ON-OFF the source. This, coupled to the nodding of the telescope, results in a three-beam technique which allows comparison between the source signal and that from two opposite empty regions of sky. The chop throw was set to be 70", which is larger than the optical diameter of the sample galaxies. Each source has been observed for 200 seconds with n depending on the expected 1.2 mm intensity. The latter has been approximately evaluated extrapolating the IRAS 100 μ m flux using a thermal spectrum with an opacity spectral index between 1 and 2. Atmospheric transmission has been monitored by frequent skydips. Uranus has been used as primary calibrator by assuming a weighted effective temperature at this wavelength of 93 ± 1 K. Several quasars have been used as secondary calibrators mainly to detect sky variations during the observations. The overall accuracy on the detected fluxes was good (± 10 %) because of the optimum atmospheric conditions. Twenty-eight objects have been observed, for 12 of which reliable fluxes (better than 3 sigma values) have been obtained. The observed values of the millimetre flux have been corrected for the overall system response and K-correction. For each source the IRAS fluxes have been taken from the IRAS Point Source Catalogue, slightly modified to account for colour and K-corrections. The Average mm to Far-IR Flux Ratio Histograms of the ratio mm to 100 μ m are reported in Figure 1. Panel (a) refers to those data without any aperture corrections. A technique of survival analysis has been used to make full use of the information content in the upper limits to the 1.25 mm fluxes. In this way we have estimated an average flux ratio of 2.02 ± 0.36 . Right-hand side panels of Figure 1 also report flux ratios derived from data collected with the IRAM 30-m antenna, whose results have already been reported in this Journal. The same analysis has been applied to the data on 26 spiral galaxies observed with the IRTF 3-m telescope. The histogram of their mm to far-IR flux ratios is plotted in Figure 2. The survival analysis provides in this case a substantially higher value for the average ratio of 1.4. An opposite indication comes from observations at 350, 450, 800 and 1100 μ m performed by Eales et al. on a few nearby galaxies. In contrast with the previous results, these authors claim that the energy distribution in the far-IR/submillimetric range is well fitted by a thermal emission of warm dust

at 30-50 K, which implies a substantially lower value for the average mm to far-IR flux ratio. Indeed, by extrapolating their model to 1.25 mm, we find in this case a value smaller by at least a factor of 2 than that implied by our observations. The discrepancy is even larger with previous results (more than a factor of 10 in this case). Some other sub-mm observations are not even commensurate with ours. Stark et al. mapped four Virgo spirals at 160 and 360 μm at a spatial resolution of $\sim 45''$, but with a poor S/N ratio at 360 μm . Thronson et al. observed only the very centre of large active galaxies at 1.3 mm, sampling a too small portion of the entire galaxy disks. Therefore, they probably lost most of the millimetric flux. To conclude, galaxy spectra in this spectral domain, hence the total amount of dust, are uncertain by up to one order of magnitude. We will briefly discuss possible origins of this large discrepancy and suggest how further investigations might be helpful in elucidating this question.

Beam-aperture corrections may contribute to explain these discrepancies. Unfortunately, the lack of knowledge of the spatial extension of galaxies at long wavelengths hampers a precise estimate of this effect. Moreover, in some ON-OFF observations, the beam separation was too small in comparison with the larger optical dimensions of the objects, and only a gradient in the emission has probably been measured.

Aperture Corrections to mm Fluxes In order to consider the whole effect of a Gaussian beam on the detected fluxes, a convolution of the beam-shape with the light profile of the galaxy millimetric emission must be done. We suppose that the millimetric light profile is exponential with scale-length λ_{mm} . We have considered the following two hypotheses: The radial distribution of cold dust closely follows that of the blue light. In this case the millimetric scale-length λ_{mm} is equal to that in blue light, λ_{opt} . The corresponding average flux ratios after correction for aperture are reported for the SEST sample and for the IRAM. Therefore, on average the detected fluxes must be corrected by a factor greater than 2. The second hypothesis considered here takes into account that the millimetric scale length is one third of the optical. In this case, we find as mean ratios: for the SEST sample and for the IRAM sample. In this case a large fraction of the millimetric flux would have been detected. From these simple considerations we have shown that the millimetric fluxes, hence the amount of cold dust in galaxies, are crucially dependent on the size of the objects

at millimetre wavelengths. The Far-IR Extension of Normal Galaxies The obvious solution to the problem of estimating galaxy broad-band spectra in the mm range will be to use arrays of bolometers, when available, to cover the entire.